

Review

Large language model for knowledge synthesis and AI-enhanced biomanufacturing

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Large language models (LLMs) are transforming synthetic biology (SynBio) education and research. In this review we cover the advancements and potential impacts of LLMs in biomanufacturing. First, we summarize recent developments and compare the capabilities of US and Chinese language models in addressing fundamental SynBio questions. Second, we discuss the application of LLMs in extracting SynBio information from unstructured data, constructing knowledge graphs, and enabling retrieval-augmented generation. Third, we anticipate that LLMs will not only revolutionize the design–build–test–learn (DBTL) cycle in metabolic modeling and engineering but also enable self-driving laboratories in future biomanufacturing. Finally, we emphasize the need for establishing benchmarks for LLMs, fostering trustworthy knowledge synthesis, developing biosecurity frameworks to prevent misuse, and encouraging collaboration among artificial intelligence (AI) scientists, SynBio researchers, and bioprocess engineers.

Highlights

Recent large language models (LLMs) are compared for their trustworthiness in synthetic biology (SynBio) studies.

Knowledge graph facilitates literature data retrieval and organizations.

LLMs empower metabolic models to gain new insights and predictions.

LLMs automate experimental designs, decision-making, and self-driving laboratories.

Rapid growth of LLMs for future biotechnology

Biomanufacturing is the use of engineered biological systems for producing valuable chemicals and pharmaceuticals. Due to advancement in synthetic biology and systems biology, the exponential growth of biotechnology is reshaping the landscape of biomanufacturing. Meanwhile, Large Language Models (LLMs) have become a groundbreaking class of generative AI that has revolutionized our research through knowledge generation, summarization, classification, and complex problem solving [1]. Recent breakthroughs have shown promise in LLMs' capabilities for human-like intelligence in assisting biological research and education. Among published LLMs, some are open source and flexible for model customization; others offer out-of-the-box solutions with proprietary optimizations. These LLMs include feature variations in task type, model architecture, parameter size, accessibility, and context length (Table 1 and File S1 in the supplemental information online). Figure 1 shows that the USA and China are the leading nations in AI research. Industries are currently playing a major role in large AI model development, but academia produces most biological models.

The rapid progress of biotechnology, fueled by the abundance of bioinformatics data, has accelerated the adoption of LLMs [2]. First, genetic and protein sequencings proliferate with datasets reaching up to 7 billion entries, facilitating **protein language models (pLMs)** (see Glossary) [3]. For example, Evo, a 7-billion-parameter genomic foundation model, can predict functions of proteins – for example, clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated protein (Cas) systems – from genome sequences at single-nucleotide resolution, and can learn how DNA mutations affect whole-organism fitness [4]. Second, LLMs (e.g., BioGPT) are developed for knowledge mining of biomedical literature [5]. Third, language models, such as LOWE (LLM-Orchestrated Workflow Engine), support drug discovery and biosynthesis of new

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compounds (www.recursion.com/lowe). Currently, LLMs and various AI approaches are rapidly co-evolved to decipher the rule of life and to design new biological systems [6]. More broadly, NASA and IBM are building a foundation model (INDUS) that spans a wide range of disciplines, including biology, physics, planetary science, and astrophysics [7]. In this review, we focus on LLMs' applications in biomanufacturing and offer a forward-looking perspective on how LLMs can drive new paradigms in **design-build-test-learn (DBTL) cycle** for cell factory development.

Biotechnology data retrieval with LLMs

Cell factory development requires multilevel engineering to optimize biomanufacturing. Researchers must not only work on gene expressions, enzyme engineering, and pathway modifications, but also need a holistic understanding of microbial responses and regulations under scale-up bioreactor conditions such as stresses, metabolic burden, genetic mutations, and oxygen limitation [8]. For training AI models, it is a necessity to have a trustworthy and complete knowledge database. Such a database can be created via literature mining of SynBio (or metabolic engineering) publications (a PubMed search resulting in over 125 000 papers, October 2024). Since structured datasets are not available in the literature [9], LLMs can automate data labeling and retrieval, which previously relied on labor-intensive manual input [5]. For instance, pre-trained LLMs can perform knowledge distillation to select features from SynBio literature, including bioreactor conditions, substrates, products, nutrient supplements, biosynthesis pathway steps, SynBio genetic components, etc. [10]. These features are influential factors to design and predict biomanufacturing performances. Various LLMs (e.g., Llama 2) have been successfully developed for feature identification in medical studies [11,12], which can also be applied for extracting biomanufacturing information from text-based files [13]. In addition, most recent GPT4o and o1 can even extract fermentation data from figures, which facilitates the calibration of bioprocess model parameters [14].

To improve the reliability, explainability, and reasoning of LLMs, knowledge graphs are used to visualize and organize SynBio information from different literature sources [15]. Furthermore, language models can themselves be employed to find similar terms in multimodal knowledge graphs, expanding their utility in synthesizing diverse datasets [16]. Human knowledge consistently fine-tunes LLM performance: techniques such as Quantized Low-rank Adaptation (QLoRA) [17]/Low-rank Adaptation (LoRA) [18] help the training process significantly by selectively finetuning a subset of LLMs' parameters. However, a small sample size can sometimes diminish an LLM's capacity to retain and apply general knowledge effectively (i.e., overfitting).

Reinforcement learning through human feedback (RLHF) [19] and **retrieval-augmented generation (RAG)** [20] have been proposed to address these challenges. Platforms such as OpenAGI [21] and SELF-ALIGN [22] leverage RLHF to enhance task-solving abilities with minimal human supervision while maintaining research integrity. The RAG framework combines a retriever model to gather external documents and custom data for improving information retrieval [7].

Comparison of trustworthiness of LLMs for SynBio Q&A

We developed a Question and Answer system (SynBioGPT) to test open-source LLMs performance by sourcing its database from the literature, including 51 777 abstracts and 23 318 open-access full-text PDFs. The training process involves one NVIDIA A40, 40G; during training, each model takes up around 20G. Utilizing LangChain for content segmentation and OpenSearch for indexing, we embedded chunks as `knn_vector` for search functionality. Through testing with 100 high-quality Q&A pairs, we optimized parameters to a chunk size of 300 and Top N of 8, finetuned LLM parameters, and applied prompt engineering with a temperature coefficient of 0.01 to maintain consistent outputs. Our evaluation included 79 specific and 21 open-ended questions covering various SynBio topics, such as gene mutation, product queries, and

Glossary

Active learning (AL): a machine learning approach where the model selectively queries the most informative data points for labeling to improve its performance efficiently with minimal labeled data.

Cell-free protein synthesis (CFPS): an *in vitro* platform that enables rapid protein production using cellular extracts without the need for living cells.

Design-build-test-learn (DBTL) cycle: an iterative process in synthetic biology and bioengineering where designs are constructed, tested, and analyzed to inform subsequent improvements.

Knowledge synthesis: the method that extracts scientific information from identifying evidence, analyzing data, and combining them to form a collected knowledge base on a topic.

Protein language model (pLM): a model that, instead of letters, inputs amino acids to a natural language model to predict protein structure and function.

Reinforcement learning through human feedback (RLHF): a method that uses human-labeled data to guide the reinforcement learning process, improving model behavior based on real-world feedback.

Retrieval-augmented generation (RAG): a method that enhances generative models by incorporating information retrieved from external databases or documents.

Self-driving laboratory (SDL): an AI-powered platform with a self-accelerating nature that uses robotics and other hardware to automate experiments and data collection.

Simplified molecular input line entry system (SMILES): a notation that encodes a molecule's 3D structure using a string of characters that can be understood by computer software (e.g., ethanol: C-C-O).

Table 1. Selected off-the-shelf LLMs developed worldwide^a

Model name (country)	Sizes (context window) ^b	Description	Time released by company	Accessibility
ChatGPT o-series ^c (USA)	o1 (200k), o1-mini (128k), o3-mini (200k)	OpenAI's reasoning models designed for step-by-step logical problem-solving	Jan 2025 by OpenAI	API, web interface
Qwen 2.5 (China)	base, VL, 1M, Coder, Math	LLM with both multimodal and multilingual abilities	Jan 2025 by Alibaba	API, web interface, Alibabacloud
Deepseek (China)	V3 (128k), R1 (128k) and distilled models from Qwen and LLaMA	R1 is the reasoning model trained through large-scale reinforcement learning	Jan 2025 by deepseek-ai	API, web interface
LLaMA 3.3 (USA)	70B (128k)	A collection of open-source decoder-only LLMs, pre-trained on trillions of tokens	Dec 2024 by Meta/Facebook	API
Mistral ^c (France)	Large-2-123B (128k), Nemo-12B (128k)	Specialized state-of-the-art models for coding, math, research, etc.	Jul 2024 by Mistral AI	API, web interface
ChatGLM 4 (China)	9B (8k), 9B-chat (128k), 9B-chat-1M (1M), 4V-9B (8k)	A family of decoder-only LLMs optimized for Chinese dialogue; 4V is the multimodal version	Jun 2024 by Zhipu AI	API
Falcon 2 (UAE)	11B (8k)	A family of causal decoder-only models and trained on RefinedWeb dataset. VLM is the vision adapted version	May 2024 by TII UAE	API
ChatGPT 4o ^c (USA)	~2T (128k)	General LLMs with tasks like Q&A, generation, etc., reading multimodal files as input	May 2024 by OpenAI	API, web interface
Yi 1.5 (China)	34B, 9B, and 6B, each with 4k, 16k, and 32k context length	A series of decoder-only LLMs pretrained on a high-quality corpus	May 2024 by 01 AI	API, web interface
Gemini 1.5 Pro ^c (USA)	NA/ (2M)	A multimodal LLM for long context text, video & audio	May 2024 by Google DeepMind	Web interface, Google cloud
Cohere Command R+ ^c (Canada)	104B (128k)	LLM designed for conversational agents, summarization, and copywriting. Compatible with their RAG model	Apr 2024 by Cohere for AI (C4AI)	API, web interface
Claude 3 Opus ^c (USA)	2T (200k)	Multimodal encoder-decoder models with Opus being the largest	Mar 2024 by Anthropic	API, web interface, Google cloud
Gemma 2 (USA)	9B (8k), 27B (8k)	A collection of open source and light weight version of Gemini built through distillation from larger models	Feb 2024 by Google DeepMind	API, Google cloud
Orca 2 (USA)	7B (4k), 13B (4k)	A finetuned version of Llama2 for research purposes only and may not behave well for specialized task.	Nov 2023 by Microsoft	Inference Endpoints
Ernie Bot 4.0 ^c (China)	260B (8k)	Model with five Chinese natural language processing tasks including natural language inference, semantic similarity, named entity recognition, sentiment analysis, and question answering	Nov 2023 by Baidu	API, web interface
Vicuna 1.5 (USA)	7B, 13B, each with 4k, 16k context length	Model trained by finetuning Llama2 on user-shared conversations	Aug 2023 by LMSYS Org	API
PaLM 2 ^c (USA)	340B (8k)	A dense decoder-only model trained with the Pathways System enables efficient training of single model across multiple Tensor Processing Units	May 2023 by Google DeepMind	API, Google cloud
Jurassic 2 ^c (Israel)	7B, 17B, 60B (8k for selected instances)	A multilingual decoder-only model with enhanced performance in instruction following, due to comprehensive instruction tuning on proprietary data	Mar 2023 by AI21 Studio	API, web interface

Table 1. (continued)

Model name (country)	Sizes (context window) ^b	Description	Time released by company	Accessibility
Luminous ^c (Germany)	13B (2k), 30B (2k), 70B (2k)	A multimodal LLM that can work with images. Trained on five languages including English, German, French, Italian, and Spanish	Feb 2023 by Aphel-alpha	API, web interface
Bloom (multinational)	1B1 (2k), 1B7 (2k), 3B (2k), 7B1 (2k), 176B (2k), 560B (2k)	A multilingual causal decoder-only LLM modified from Megatron-LM	Jul 2022 by BigScience	API

^aThe table includes major LLMs developed up until January 2025.

^bInput token length: 1 k, 1 thousand; 1 M, 1 million; 1 B, 1 billion; 1 T, 1 trillion.

^cClosed source/partially open LLMs.

competitive pathway exploration (File S2 in the supplementary information online). Among eight LLMs (Figure 2A), Llama3 showed the best accuracy. After RAG by using abstracts of SynBio papers, the accuracy of all LLMs greatly improved from 25% to >65%. Further RAG from the full text of the papers, Qwen from Alibaba showed similar accuracy to Llama3 (~85%). Based on our recent generative AI study [10], Figure 2B compared Qwen with the OpenAI GPT4 for their capability to retrieve data from seven papers on yeast fermentation. In general, GPT4 can obtain more tabular data than Qwen (containing only 14B parameters). When the number of model parameters increases to 72B (i.e., use a bit more computational resources), Qwen can obtain about the same amount of data as GPT-4. However, two further AI evaluation metrics were employed: Recall-Oriented Understudy for Gisting Evaluation (ROUGEL) and Bilingual Evaluation Understudy (BLEU), respectively (Figure 2C,D). Both evaluation methods showed that RAG improved our LLMs performance to tackle complex problems. Moreover, ROUGEL scored much better than BLEU, indicating that RAG effectively enhances LLMs' capability for extracting contextually relevant information. However, RAG still shows limitations in improving the precision of the retrieved data.

LLM as a biological sequence model

Sequencing data (DNA/RNA/protein) from functional genomics studies have a natural affinity for language models, as they have letters as building blocks. Recently, AI and omics approaches were integrated to simulate the diverse behaviors of molecules and cells, offering groundbreaking opportunities to create an AI virtual cell (AIVC) [23]. Specifically, the first layer of the AIVC focuses on modeling DNA, RNA, and proteins as character sequences of nucleotides or amino acids, forming a universal representation (UR) on the molecular level. Compared with the traditional sequence-predicting deep learning (DL) model [24], LLMs have a more unified structure and can be easily adapted without substantial task-specific data. This adaptability is due to their pre-training and finetuning processes, which allow LLMs to offer multimodal and end-to-end solutions for extracting or generating target sequences. The LLM-based sequence models include ProGen [25,26], xTrimoPGLM [27], ProGPT2 [28], and ProLLaMA [29], which has shown superior capabilities in protein property prediction and structural optimization. Notably, OpenCRISPR-1 [30] fine-tunes the ProGen-based model with CRISPR-Cas Atlas to generate diverse CRISPR-Cas proteins with enhanced activity and specificity. Moreover, Nvidia's BioNeMo biomolecular language model has combined state-of-the-art AI modeling tools (e.g., AlphaFold) to precisely predict structures of over 40 000 human proteins [6].

Besides pLMs, gene expression is another form of biological language that determines metabolic regulations and cell functions. Recent studies created pretraining single-cell foundation models: scFoundation for gene expression and drug response prediction [31] and scGPT for multiomics data analysis [32]. Moreover, scMulan uses a multitask generative pre-trained language model to

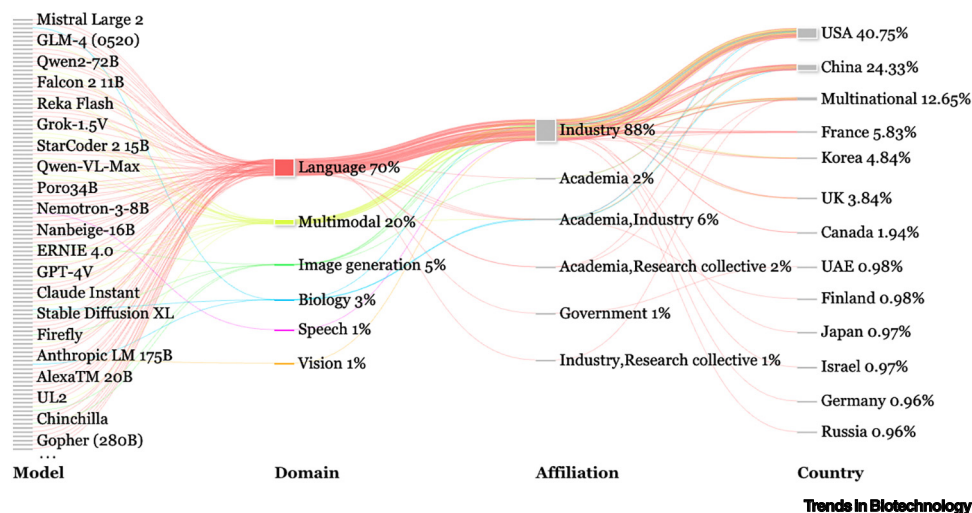


Figure 1. Sankey graph comparing the top 100 large-scale artificial intelligence (AI) models (requires more than 10^{23} floating point operations to train) released between 2020 and 2024 (File S1 in the supplemental information online). It summarizes the domains and affiliations of multimodal large language models (LLMs). The percentage of models is shown for each category in the corresponding column. The order of model is according to publication date (newest to oldest).

decode cellular functions from single-cell transcriptomic data [33]. Current sequence language models lack tokenizers specifically tailored for sequencing data that can effectively capture and interpret its biological significance. The integration of graph neural networks and sequencing data (graph sequence models) shows great potential to advance genome-to-phenome predictions [34,35].

LLM-powered cell factory development

Cell factory development requires enzyme engineering, pathway design, and fermentation optimizations. Due to complex metabolic regulations, biomanufacturing development relies on a DBTL cycle in which strain development is guided by both model predictions and experimental tests. We envision that the development of LLMs and microbial cell factories via DBTL have mutual benefits (Figure 3).

LLMs can guide microbial cell factory development. Through textual transformation and feature selection from literature and experimental reports, LLMs accelerate machine learning (ML) workflows to predict cell performance under bioreactor conditions. Moreover, features of traditional ML models are often not sufficient to describe multilevel biological properties [36]. LLMs, as knowledge extractors, can identify and rank hidden influential factors and make ML more explainable [37]. These influential features and the corresponding feature scores can be used to guide fermentation optimizations and reduce experimental costs. Moreover, LLMs can offer effective strategies for strain engineering. A recent study demonstrated that LLMs could enhance DBTL cycles by extracting large-scale metabolic engineering methods from over 29 000 entries, spanning 1210 products and 751 organisms [38]. Another study showed that the LLM (Qwen) with RAG could not only provide genetic targets and experimental design for *Yarrowia* yeast pathway engineering but also lead to a new hypothesis for a biosynthesis pathway [15].

DBTL cycles require reliable metabolic predictions. However, LLMs would fail at quantitative objectives, such as predicting metabolite productions under complex process conditions, due to poor reasoning in math [39]. Therefore, LLMs can integrate with white box models such as the

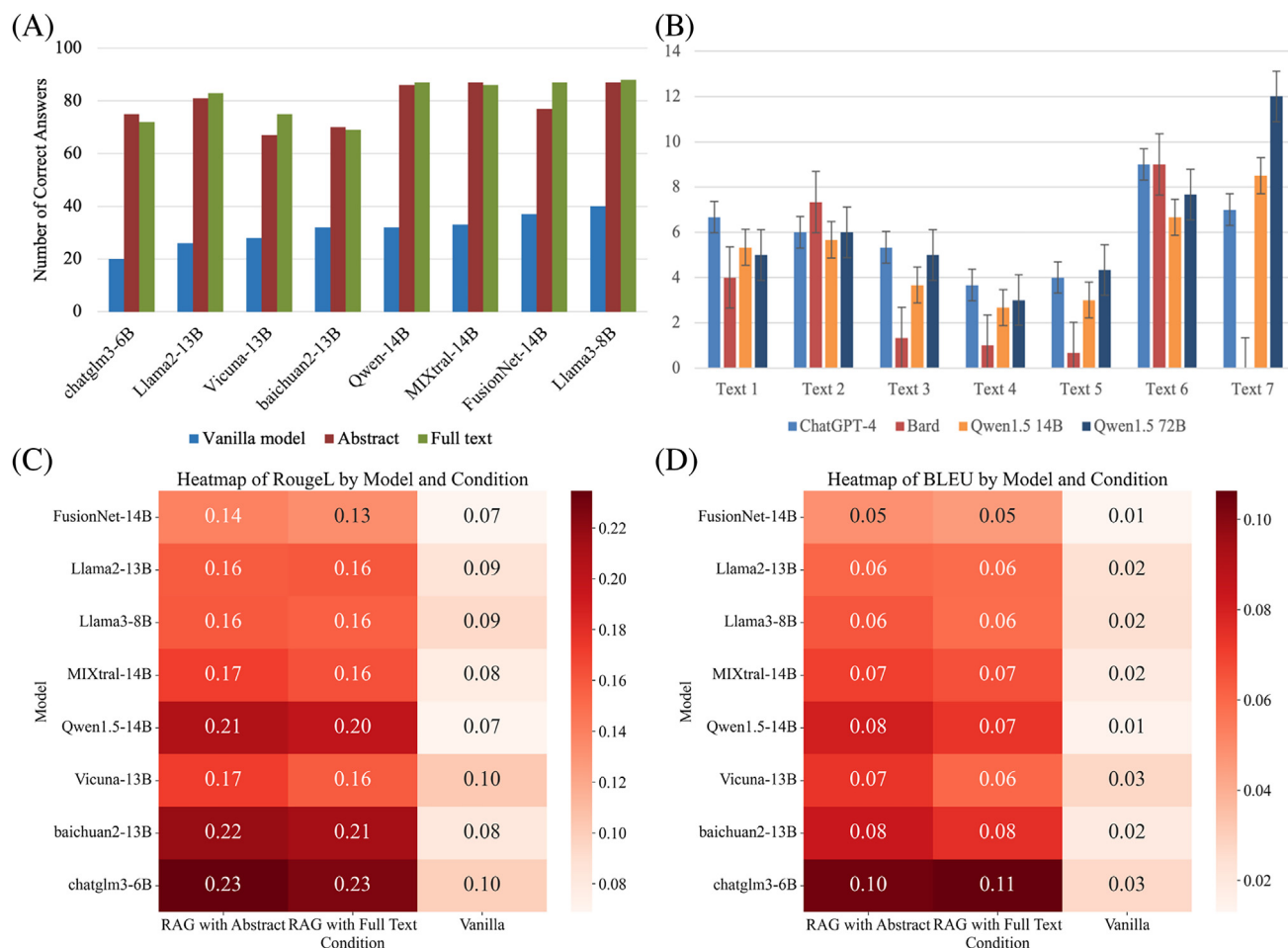


Figure 2. Comparison of large language models (LLMs) in synthetic biology (SynBio) tasks (File S2 in the supplementary information online). Vanilla means using the default model without any enhancement. (A) Distribution of correct answers given by eight models under three conditions: using original model (Vanilla model), retrieval-augmented generation (RAG)-enhanced with abstract of the papers collected (Abstract), and RAG-enhanced with full text of the papers (Full text). (B) ChatGPT-4 compared with smaller models Qwen1.5-14B and Qwen1.5-72B in extracting tabular data from seven different SynBio papers (labeled Text 1–7). Each model was tested three times, and the error bar was the resulting standard deviations. The y-axis refers to the number of tabular data points generated. (C) The heatmap displays the average ROUGE L (Recall-Oriented Understudy for Gisting Evaluation) scores of LLMs (y-axis) under RAG enhancement conditions (x-axis). (D) The heatmap displays the average BLEU scores of LLMs under RAG enhancement conditions with a maximum subsequence length of 4. Note: ROUGE L evaluates how much reference is covered and BLEU evaluates the closeness/accuracy of answers to the reference.

Genome-Scale Model (GSM) [36,40] and Techno-Economic Analysis (TEA) [41], which can expand LLM reasoning and aid commercial decision-making for biomanufacturing. However, mechanistic models make simplified assumptions, while LLMs can bridge the gap between model assumptions and real-world conditions [42]. Specifically, protein/RNA/DNA-based ‘language’ can be incorporated into multiscale omics data and GSM for improved phenotype understanding and biological quantification [43,44]. For example, the knowledge of enzymatic turnover rates, derived from DL of protein sequences and/or substrate structures, facilitates metabolic modeling of industrial CHO cell lines [45].

The integration of LLMs with AI and omics data is poised to drive the evolution of genome-scale modeling into whole-cell modeling, enabling the analysis of interactive genetic and metabolic

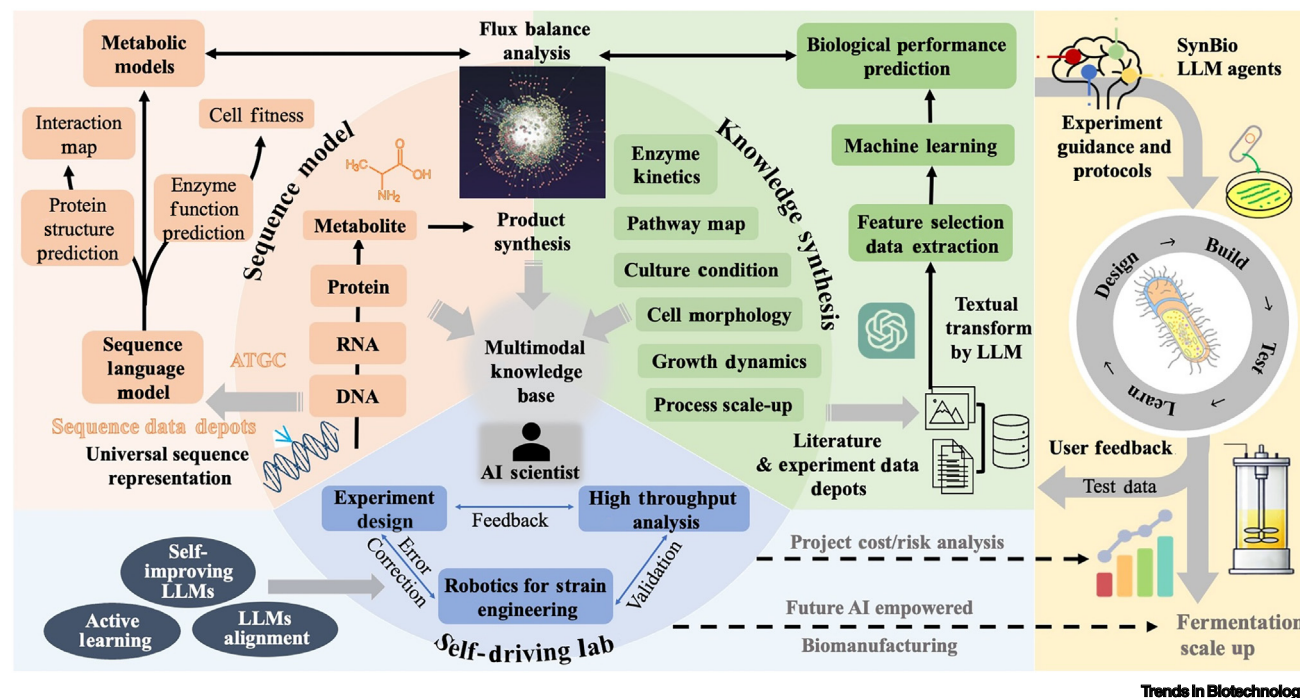


Figure 3. Large language models (LLMs) streamline the design–build–test–learn (DBTL) process of strain development into a unified and efficient pipeline. Module 1 (orange): sequence models. Module 2 (green): multimodal capabilities of LLMs for mining diverse data types, including text, tables, and graphs. Module 3 (yellow): DBTL cycle. Module 4 (blue): self-driving module, which includes LLM and active learning (AL) for experiment design and task scheduling, automating labor-intensive processes with robotic strain engineering, and quality control through human supervision.

functions and the prediction of nonlinear dynamic responses in microbial cell factories [23]. These advanced simulations, often referred to as AI virtual cells, are expected to revolutionize *in silico* cell engineering experiments and significantly enhance the efficiency of the DBTL cycle.

Self-driving laboratories with LLM agents

Although DBTL is a current cell factory development strategy, we envision that LLMs will serve as the intelligent agent to empower biomanufacturing by further mitigating the labor-intensive hardware-enabled research. Specifically, ‘AI scientist’ is a new intelligent system that utilizes LLMs to help human scientists with large-scale analysis and repetitive DBTL task execution [46]. This workflow leads to a paradigm shift toward the **self-driving laboratory (SDL)**, including intent (converting a SynBio objective into a computational workflow), decision (making decisions on experiment methods), execution (scheduling and monitoring workflow), and analysis (materializing workflow outputs) [47]. The synergy between LLMs, robotics, and human oversight facilitates rational design, ultimately culminating in biofoundry design and fermentation scale-up.

SDL has the following advantages. First, LLMs assist task planning by breaking a complex task into chains of sub-reasoning steps [39]. For example, ChemCROW utilizes a chain-of-thought reasoning loop to break down complex tasks and identify relevant tools to interact with the physical world [48]. Methods such as Decompositional Distillation [49] can divide the pathway optimization objective into enzyme selection, pathway construction, and flux analysis, which can be individually resolved by specific tools. Second, LLMs automate experimental design and planning. For example, Self-driving Autonomous Machines for Protein Landscape Exploration (SAMPLE) [50] learn sequence–function relationships to design desired proteins, while **simplified molecular**

input line entry system (SMILES) translates a chemical's structure into a string of symbols that supports the design of enzyme reactions [51]. Moreover, LLMs can support information organization, subtask reasoning, selection of suitable tools for complex tasks, experiment protocol preparation, and data analysis as demonstrated by ChemCROW [48] and Coscientist [52]. Recently, CRISPR-GPT has been augmented with domain knowledge and external tools to design gene-editing experiments [53]. Third, an LLM-based knowledge retriever can be integrated with cloud computing and hardware control to give more flexibility and scalability to automate laboratory tests [54]. For example, SynThreader was originally developed for retrieving experimental protocols via natural language processing, which can be integrated with LLMs to significantly improve protocol extraction efficiency and accuracy [55]. Moreover, LLMs can reduce human intervention by predicting and monitoring experimental constraints and outcomes. For example, LLMs may direct media preparation, pH adjustments, and other routine biomanufacturing work during fermentation processes [56].

Meanwhile, advancements in AI facilitate better SDL (Figure 3). **Active learning (AL)** prioritizes biological experiments by identifying the most impactful data points to collect or validate. Recent advancements showcase how GPT-4 and AL optimize **cell-free protein synthesis (CFPS)** with cluster margin selection to explore experimental conditions and improve antimicrobial protein yields [46,57]. Moreover, self-improving LLMs can generate increasingly precise hypotheses for experiments by looking at their own reasoning to detect mistakes [58]. Lastly, data security has been a major concern for biomedical applications. LLM alignment ensures that models act in accordance with both ethical/security considerations and scientific rigor [59].

Biosecurity for AI-biotechnology integration

Current biosafety standards in the AI-biotechnology fusion are not enough. On one hand, LLMs have increased the potential for bioweapon or pathogen synthesis [60]. Previously, various regulations controlled the access to dangerous and sensitive knowledge. If LLMs find cures for pathogens and cancers [61], they could also make new and more harmful pathogens for biological attacks [62]. For example, LLMs can inform about variola virus, engineered influenza, *Bacillus anthracis*, and *Yersinia pestis*. LLMs can assess the feasibility, time, cost, and logistical aspects of acquiring and distributing biological agents [63]. Individuals can acquire infectious virus samples using methods such as reverse genetics based on instructions from LLMs. With indications of beneficial intent, LLMs may bypass security measures, proving the current safety protocol inadequate [64]. It becomes a race against time to see who finishes faster: individuals who materialize them in the laboratory, or the safety experts who establish working procedures. On the other hand, LLM regulations are urgently needed to ensure data privacy and intellectual property confidentiality [65]. To avoid information breaches, the use of a secure LLM application programming interface is a necessity. Other approaches include deploying LLMs locally, employing privacy-preserving techniques, zero-knowledge proofs, and federated learning to protect sensitive information during LLM training and deployment [66]. In addition, the recent White House Executive Order on AI [67] mandates that developers training AI models, especially for SynBio, involving more than 10^{23} computational operations report to the federal government regarding their cybersecurity, red-teaming, and other security measures. Nevertheless, for the benefit of the scientific community, we should focus on harnessing technological advancements responsibly and constructively, rather than attempting to halt their progress.

Challenges and future directions

LLMs are not yet widely used in biomanufacturing industries. We envision the following challenges. One key roadblock in generative AI applications is the lack of pipelines to seamlessly

integrate multimodal and unstructured data from different bioinformatics resources [36,68]. SynBio research generates various types of knowledge, such as images (metabolic pathway diagrams, molecular structure [69–71], cell morphology images [72], etc.) and tables (mass spectra [73], multiomics expression data [74], etc.). Particularly, image-to-text tasks in literature mining have been studied in areas such as named entity recognition (NER) [75,76] and pathway extraction [77]. However, traditional DL-based image extraction methods often suffer from subpar accuracy due to various challenges, such as inconsistent chemical terminology [77] and the need for manual validation of results [10]. Besides, extracting time series data from literature figures or plots remains a challenge, as current LLMs can only identify data trends [78,79]. Therefore, we envision that future LLMs may address these challenges by comprehending SynBio terminology, bridging varied image formats, automating image-to-data organization, and standardizing time series data.

Biological systems are highly complex and poorly reproducible. AI approaches may suffer from an incomplete knowledge base content, information biases, and fragmented or distracting information. In particular, the majority of data in SynBio literature is in the field of strain engineering, laboratory cultures, and molecule characterizations. However, very few scientific papers focused on biomanufacturing process scale-up or strain stability over long growth generations. Thereby, the data collection from pilot studies and fermentation industries (i.e., high Technology Readiness Levels >3) is a necessity to ensure quality and commercial applicability of LLMs. Future SynBio database construction and **knowledge synthesis** could benefit from the following LLM methods: (i) to reconcile conflicts [80] and reject false information [81], LLMs have shown great potential to filter and standardize data through self-guided evaluation of instruction data [82]; (ii) LLMs can connect and highlight SynBio knowledge nodes and sources via graph network [15]; (iii) optimization of long prompts for LLMs to ensure meaningful representation [83]; and (iv) LLMs perform better when exposed to diverse training data [84], thus, sharing of laboratory and large-scale fermentation data among academia and industries can fine-tune LLMs' applicability. This finetuning process involves curating high-quality, diverse datasets from both laboratory experiments and industrial-scale operations, followed by domain-specific pretraining and task-specific supervised learning [85]. AL techniques, where models iteratively improve through feedback from domain experts, can further enhance LLM trustworthiness [57].

The question of how to test LLMs remains open, particularly for reasoning and fact-checking-related tasks. Various benchmarks [86–89] have emerged to evaluate LLMs on different levels of intelligence. However, man-made benchmarks are not enough for evaluating machine intelligence since LLMs sometimes 'cheat' by memorizing repeated patterns, thus failing to recognize logically preserved meaning [39], or by having seen leaked metadata information during training [84]. As LLMs continue to evolve, benchmarks must remain dynamic to effectively measure their performance. Attempts have also been made to address these challenges: diving into the mechanism for how LLMs answer questions [39], testing ability of LLMs to recognize paraphrases [89], or using an entropy-based uncertainty estimator [81] may overcome the statistical tricks LLMs sometimes throw to bypass commonsense human-like reasoning.

Concluding remarks

Current research and education aspects in SynBio and biomanufacturing are being reshaped by LLM literacy [90] and protein language model [91]. LLMs for biomanufacturing require extensive computational costs, multimodal data resources, automation facilities, domain expert feedback, and general rules of security regulations (see [Outstanding questions](#)). Academia and industry can complement each other in advancing biotech LLMs, and future research calls for global and

Outstanding questions

How can LLMs be designed to integrate biological knowledge across scales from molecular interactions to whole organisms?

With the vast amount of omics data, can LLMs genuinely capture the intricacies of complex metabolic regulations?

How can we ensure the development of explainable AI to derive interpretable results from black-box models?

Can self-driving laboratories eventually replace the traditional design–build–test–learn cycle?

multidisciplinary collaborations for driving new computing advancements to benefit scientific learning and discoveries through a virtuous cycle.

Author contributions

H.M. and Y.J.T. proposed review outlines. W.L., Z.M., and Z.X. drafted the manuscript. Z.M. and X.L. developed SynBioGPT and tested different LLMs. All authors approved the paper.

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Declaration of interests

The authors declare no conflicts of interest.

Supplemental information

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